

Introduction to Machine Learning
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University of Science and Technology of China

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Homework 3
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Notice, to get the full credits, please present your solutions step by step.

Exercise 1: Affine Sets and Projections

- (1) Let $A \in \mathbb{R}^{m \times n}$, $b \in \mathbb{R}^m$, and consider

$$U = \{x \in \mathbb{R}^n : Ax = b\}.$$

- (a) Show that U is an affine set. Identify its direction space in terms of A .
(b) Prove that for any $x_0 \in U$,

$$U = x_0 + \mathcal{N}(A),$$

where $\mathcal{N}(A)$ denotes the null space of A .

- (2) Suppose $A \in \mathbb{R}^{m \times n}$ has full row rank and $U = \{x : Ax = b\}$ is nonempty.

- (a) Consider the projection of a point $y \in \mathbb{R}^n$ onto U :

$$p = \underset{x \in U}{\mathbf{argmin}} \|x - y\|_2^2.$$

Use Lagrange multipliers to derive the formula

$$p = y - A^\top (AA^\top)^{-1} (Ay - b).$$

- (b) In $\mathbb{R}^3$, let

$$U = \{x \in \mathbb{R}^3 : x_1 + 2x_2 - x_3 = 3\}, \quad y = (1, 0, 0)^\top.$$

Compute explicitly the projection p of y onto U .

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Exercise 2: Convex Sets, Images and Preimages

Let $C \subseteq \mathbb{R}^n$ be a nonempty convex set and let $A \in \mathbb{R}^{m \times n}$, $a \in \mathbb{R}^m$.

- (1) Show that the affine image of C ,

$$D = \{y \in \mathbb{R}^m : y = Ax + a, x \in C\},$$

is convex.

- (2) Define the affine preimage of C by

$$E = \{y \in \mathbb{R}^m : Ay + a \in C\}.$$

Show that E is convex.

- (3) (Epigraph and hypograph)

- (a) Let $f : \mathbb{R}^n \rightarrow \mathbb{R}$ be a convex function. The epigraph of f is

$$\mathbf{epi} f = \{(x, t) \in \mathbb{R}^{n+1} : t \geq f(x)\}.$$

Show that $\mathbf{epi} f$ is a convex set.

- (b) Let $g : \mathbb{R}^n \rightarrow \mathbb{R}$ be concave. Its hypograph is

$$\mathbf{hypo} g = \{(x, t) \in \mathbb{R}^{n+1} : t \leq g(x)\}.$$

Show that $\mathbf{hypo} g$ is convex.

- (4) (Interior and relative interior – computation) You do not need to prove your answers, but you should explain them briefly.

- (a) Find $\mathbf{int} C$ and $\mathbf{relint} C$ for

$$C = \{x \in \mathbb{R}^3 : x_1^2 + x_2^2 < 4, x_3 \geq 0\}.$$

- (b) Let

$$K = \{x \in \mathbb{R}^n : x_i \geq 0, \sum_{i=1}^n x_i = 1\}$$

be the standard probability simplex. Determine $\mathbf{int} K$ (as a subset of $\mathbb{R}^n$) and $\mathbf{relint} K$.

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Exercise 3: Cones and Farkas-type Results

Let $A = (a_1, \dots, a_n) \in \mathbb{R}^{m \times n}$ and consider the conic hull

$$\text{cone}(A) = \left\{ \sum_{i=1}^n \alpha_i a_i : \alpha_i \geq 0 \right\} \subseteq \mathbb{R}^m.$$

- (1) Show that $\text{cone}(A)$ is a convex cone, i.e.,

$$x, y \in \text{cone}(A), \lambda, \mu \geq 0 \Rightarrow \lambda x + \mu y \in \text{cone}(A).$$

- (2) Show that $\text{cone}(A)$ is closed.

- (3) (Gordan's theorem) Let $A \in \mathbb{R}^{m \times n}$. Prove that exactly one of the following two systems has a solution:

- There exists $x \in \mathbb{R}^n$ such that $Ax > 0$ (componentwise).
- There exists $y \in \mathbb{R}^m$, $y \geq 0$, $y \neq 0$, such that $A^\top y = 0$.

Hint: Consider appropriate cones and use separation theorems (you may assume standard separation results for closed convex cones).

- (4) Show how Farkas' Lemma can be deduced from Gordan's theorem in part (3).